



Scholars High School Physics 1: Mechanics (4926)

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Fri

Sep 11, 2026 - Mar 12, 2027

7:30 PM - 9:00 PM ET (4:30 PM - 6:00 PM PT)

Homework

Week: 1 **2** 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

Homework: Week 2



Readings

You have completed **8** of **14** challenge problems.

Due **Sep 25**.

Week 2 Transcript: [Fri, Sep 18](#)

Challenge Problems

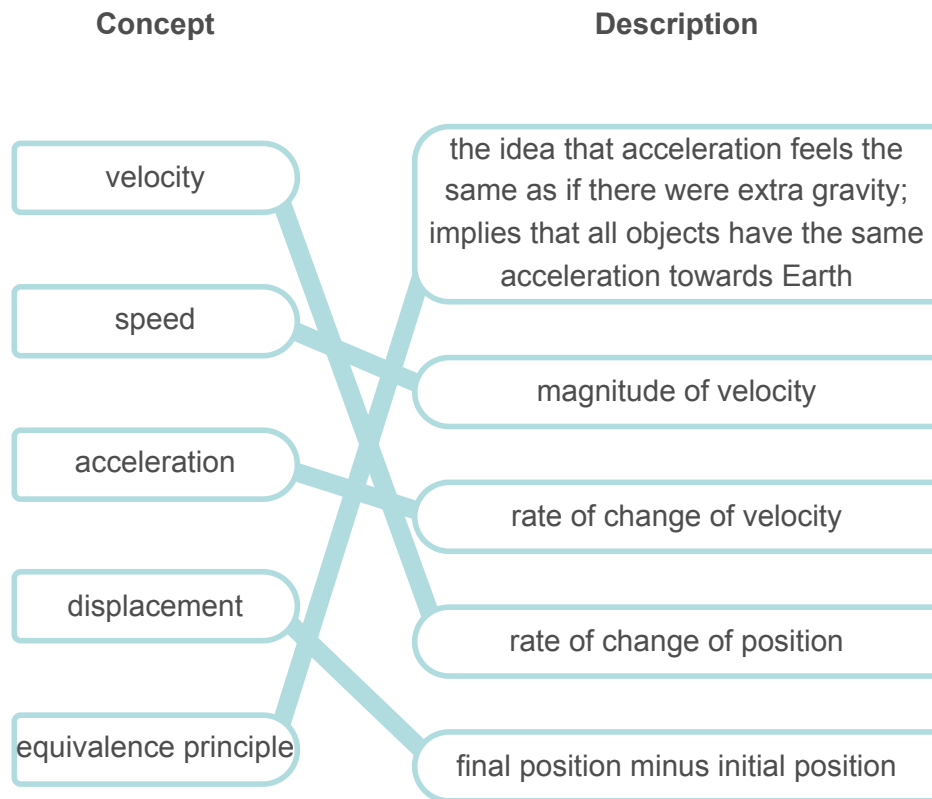
Problem:[Report Error](#)

The first three problems on this set are quick reviews to make sure you remember the terminology introduced in class. They are quick problems, and after those three, you'll dive into applying the physics to real situations.

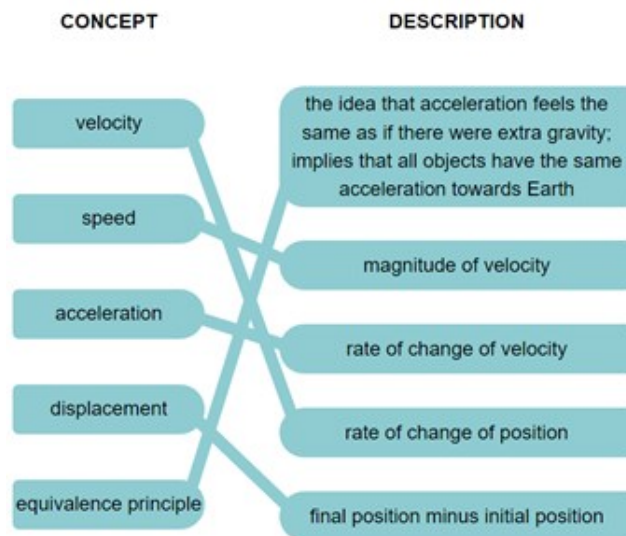
If you are unsure on these, you could check the transcripts (especially the summaries at the end of class) for weeks 1 and 2.

Match the concepts to their descriptions.

The matching is one-to-one, meaning each concept should connect to one description, and each description should connect to one concept.

**Solution:**

The connections are



These are all definitions introduced over the previous two weeks of class.

Your Response(s):



velocity: rate of change of position

speed: magnitude of velocity

acceleration: rate of change of velocity

displacement: final position minus initial position

equivalence principle: the idea that acceleration feels the same as if there were extra gravity; implies that all objects have the same acceleration towards Earth

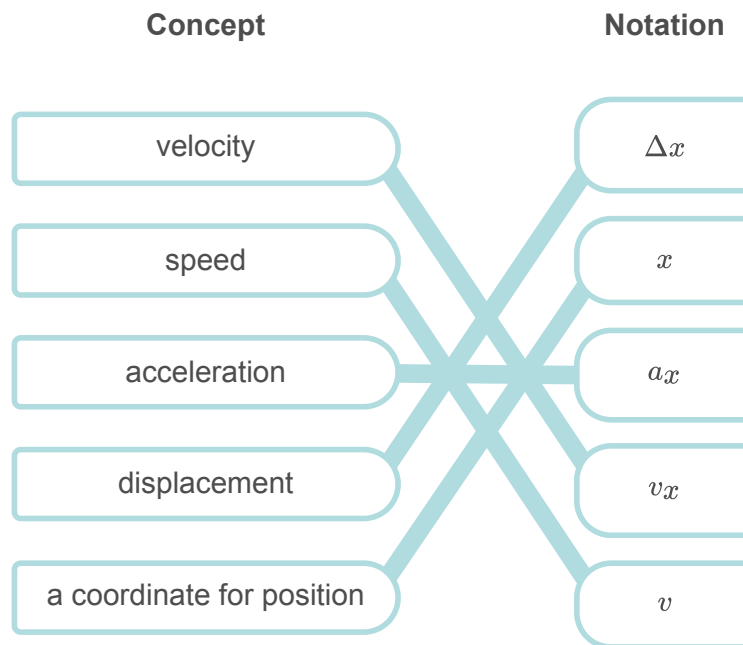
Problem 2 – Correct! – Score: 5 / 7 (46176)

**Problem:**

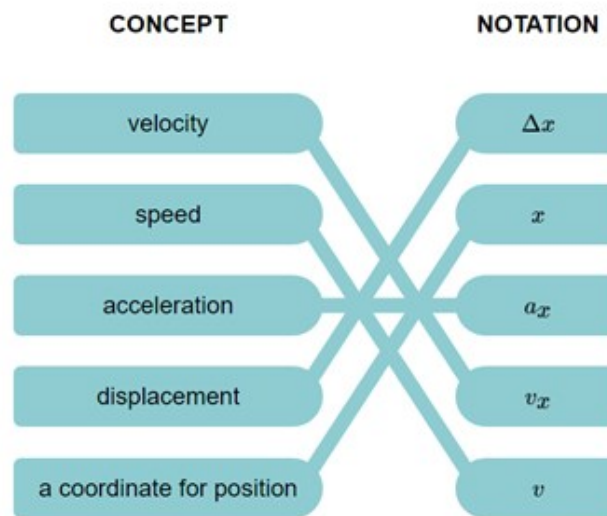
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Match the concept to our mathematical notation for it. Assume we are using x as a coordinate.

The matching is one-to-one.

**Solution:**

The connections are



These are bits of notation introduced over the previous two weeks.

Recall that v is our notation for *speed*, the magnitude of velocity. When we're dealing with velocity along a particular coordinate, for example the x coordinate, we write it as v_x . This velocity can be positive or negative depending on the direction of motion. The speed can only be positive.

Also remember that there can be other coordinates besides x . For example, for vertical motion, we'll typically use the coordinate y . The velocity for vertical motion would be v_y and the acceleration for vertical motion would be a_y .

Displacement is a change in position, so it uses a Δ in its notation. Δx is the change in the x coordinate, which is the displacement along the x direction. For a vertical displacement, we could use Δy .

Hint(s): Be careful with velocity and speed. Recall that when we want a quantity that keeps track of direction and can be either positive or negative, we use v_x for motion along the x direction. When we're only interested in how fast something is going, and not the direction, we use v . They are related by $v = |v_x|$. You'll need to decide which one is velocity and which is speed.

Your Response(s):

👎 velocity:

$$v$$

speed:

$$x$$

acceleration:

$$a_x$$

displacement:

$$\Delta x$$

a coordinate for position:

$$v_x$$

👍 velocity:

$$v_x$$

speed:

$$v$$

acceleration:

$$a_x$$

displacement:

$$\Delta x$$

a coordinate for position:

$$x$$

Problem 3 – Correct! – Score: 7 / 7 (45992)

**Problem:**[Report Error](#)

An object is moving, and we use the coordinate z to track its motion.

Create equations for the displacement, velocity, and acceleration of the object. Each should use at least one Δ .

$$\text{displacement} = \frac{\Delta z}{1}$$

$$\text{velocity} = \frac{\Delta z}{\Delta t}$$

$$\text{acceleration} = \frac{\Delta v_z}{\Delta t}$$

z v_z t Δ

Solution:

The Δ means "change in." It always comes before the variable we want to track the change in. For example, Δm means "change in the mass." When we say "rate of change," we mean change in a quantity divided by change in time, by default. For example, if you're pouring water into a cup, the mass of water in the cup is increasing. The rate of change of the mass in the cup would be $\frac{\Delta m}{\Delta t}$.

The equations are

$$\text{displacement} = \Delta z$$

$$\text{velocity} = \frac{\Delta z}{\Delta t}$$

$$\text{acceleration} = \frac{\Delta v_z}{\Delta t}$$

These are the same as the definitions given in class over the last two weeks, but we have replaced the more typical coordinate x with the coordinate z . In physics, it's useful to be flexible in how you use notation. Different sources and different sub-fields of physics use notation differently, so we need to be able to adapt to those small differences.

These equations can be read as saying,

"Displacement is a change in something's position (or its coordinate)."

"Velocity is the rate of change of displacement."

"Acceleration is the rate of change of velocity."

Hint(s): The Δ means "change in." It always comes before the variable we want to track the change in. For example, Δm means "change in the mass." When we say "rate of change," we mean change in a quantity divided by change in time, by default. For example, if you're

pouring water into a cup, the mass of water in the cup is increasing. The rate of change of the mass in the cup would be $\frac{\Delta m}{\Delta t}$.

Your Response(s):

☺ $\Delta, z, \Delta, z, \Delta, t, \Delta, v_z, \Delta, t$

Problem:[Report Error](#)

This problem asks you to think qualitatively about acceleration and free fall. You'll visualize acceleration using a free falling stick.

There is a challenge that many people find surprising: hold a dollar bill vertically and invite a friend to put two fingers on either side of it, near the bottom, ready to grab the dollar bill.



Chegg, <https://www.chegg.com/homework-help/questions-and-answers/classic-trick-person-1-hangs-dollar-bill-hand-dangles-fingers-person-2-person-1-releases-b-q31489919>

Ask your friend to try to catch the dollar after you drop it, but without moving their hand down. (This is pretty fun — try it yourself with a friend or family member!)

Free fall acceleration is high enough that almost no one can react quickly enough to catch the dollar.

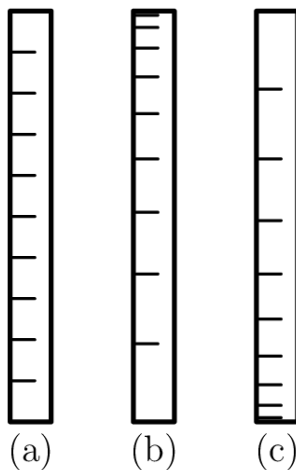
Suppose you want to modify this idea to measure your friend's reaction time. Instead of dropping a dollar, you'll drop a stick that's long enough that your friend will be able to catch it.

You know g , the local free fall acceleration, so you can calculate how the stick will fall. You put marks on the stick labeled with times in such a way that when your friend grabs it, their fingers are at a mark corresponding to their reaction time.

For example, if your friend's reaction time is 0.25 s, then when they catch the stick, their fingers will be on top of a mark on the stick labeled " 0.25 s". Your markings will correspond to reaction times of 0.05 s, 0.10 s, 0.15 s etc. starting at the bottom of the stick and increasing up to the top.

Which of the following shows the correct spacing between the marks on the stick?

You don't need to make measurements on the diagrams below. Use the general pattern of the spacings to answer this question qualitatively.



Solution:

This is stick .

The markings on the stick are very similar to a dot diagram. Each marking shows how far the stick will fall in a certain amount of time.

If your friend has a very fast reaction time, they'll grab the stick near the bottom. The slower their reaction time, the higher up the stick they'll grab it. So as reaction time increases, we move up the stick, from the bottom to the top. This is the direction we'll look at we analyze the markings.

Since the stick falls faster and faster over time, the markings should get further and further apart. They should be close together at the bottom and far apart at the top.

This is stick .

Hint(s): Over time, the ruler will fall faster and faster. That means the markings should get further and further apart, starting from the place where they first pass through your fingers.

Your Response(s):

 (c)

Problem 5 – Correct! – Score: 7 / 7 (43632)

**Problem:**

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On their wedding day, a couple jumped off a cliff into water below. (As a celebration, not out of despair.)



Zvonimir Barisin

There's another photo taken just before they hit the water. In this question, we'll try to figure out what that photo looks like. (That photo is included in the solution.) This gives us more qualitative practice with free fall acceleration.

To model the jump of bride and groom off a cliff, we make some simplifications:

- assume they both are falling vertically from the same height, but the bride started falling a moment before the groom
- ignore air drag
- ignore that they are holding hands
- model each person as a point at their *center of mass*. This is intuitively the center spot of a person. When standing, it's roughly few inches behind the belly button

Then we can define a few terms related to the gap between bride and groom:

separation: how far beneath the groom the bride is

speed gap: the difference in speed between the bride and groom

acceleration gap: the difference in magnitude of acceleration between the bride and groom

time gap: the time between the bride reaching a certain height and the groom reaching that same height

As the couple fell and before they hit the water, these quantities may have changed. **Categorize each according to whether it increased, stayed the same, or decreased.** Categories can hold multiple quantities, and categories are allowed to be empty. Every quantity needs to go in one category.

Reflecting on your answers, try to guess what the photo taken just before they hit the water looks like.

Increased	Stayed the Same	Decreased
separation	speed gap	
	acceleration gap	

**Solution:**

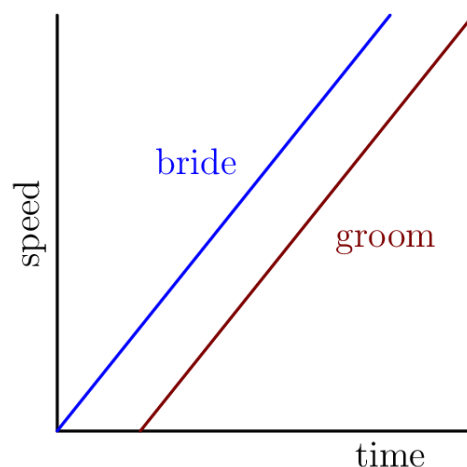
Here's the photo just before they hit the water:



The separation increased quite a bit.

To see why, we can start with the acceleration gap. Since objects in free fall all have the same acceleration, the acceleration gap is zero.

To explore the speed gap, we could draw a graph of the speed versus time for both the bride and the groom.

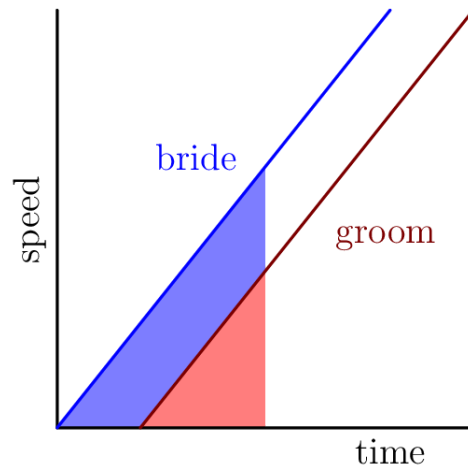


We don't need numbers on this graph because we only want to use it to guide our thinking qualitatively.

The slopes for the bride and groom are the same, since the slope is free fall acceleration. The bride's starts at the origin, but the groom's line is shifted to slightly higher times since he jumped a little later. The horizontal distance between the lines is how long the groom waited to jump.

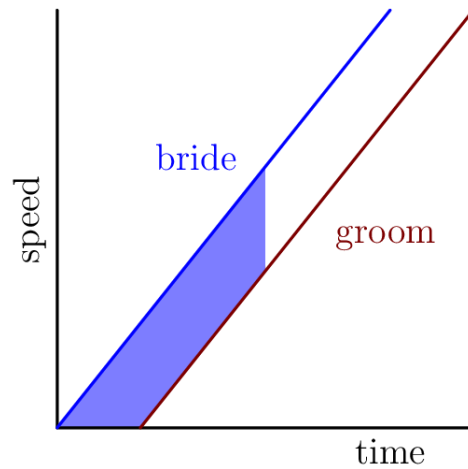
The vertical distance between the lines is the speed gap. Since the lines have the same slope, this distance doesn't change.

How about the separation? Remember that the distance fallen is the area under the velocity versus time graph. At some particular time, we can compare these two areas:

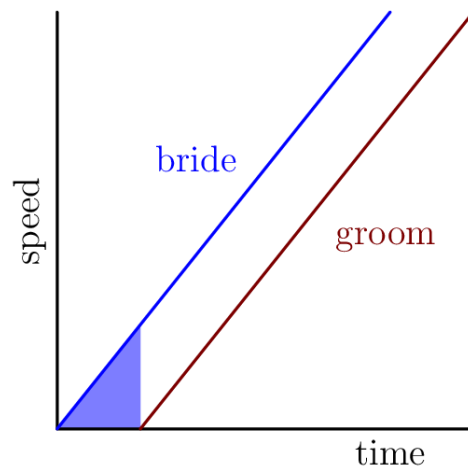


(The blue area, corresponding to how far the bride has fallen, extends back behind the red area for how far the groom has fallen.)

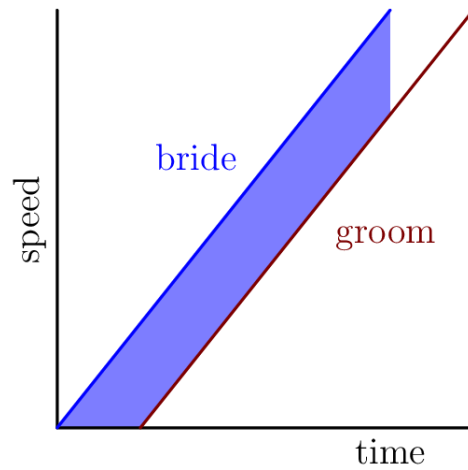
The difference in these areas is how much further the bride has fallen, so we can visualize the separation like this:



But the separation right when the groom jumped was much smaller:



And the further along in time we go, the bigger the separation gets.



Another way to think of it is to realize that, because she jumped first, the bride already has some downward velocity when the groom starts falling. The *difference* between those velocities stays the same, so the bride is always falling a little faster than the groom. Since she's falling faster, she gets continually further away.

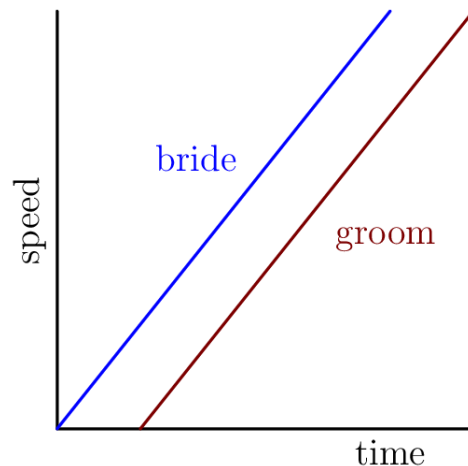
Finally, the time gap remains constant. Because they have the same acceleration and same initial speed (i.e. zero), the bride and groom fall in exactly the same way, just with the groom a moment behind, so the time gap is constant.

The correct answers are:

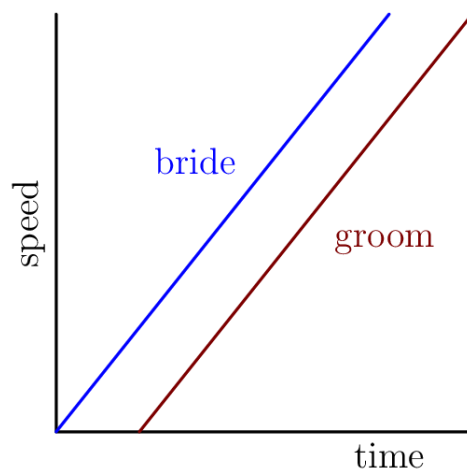
INCREASES	STAYS THE SAME	DECREASES
separation	speed gap	
	acceleration gap	
	time gap	

Hint(s): 1

Here's a plot of the speed versus time for both bride and groom.

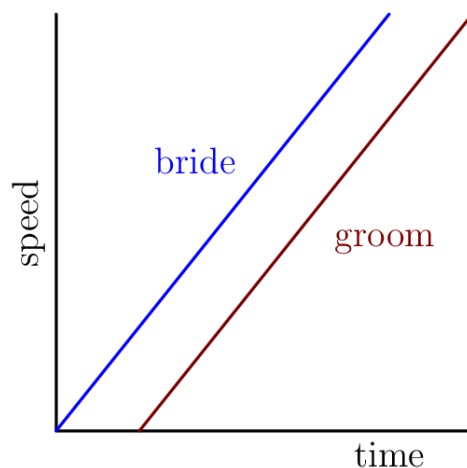


The speed gap is the vertical distance between the lines. Two speed gaps at different times are shown as Δv_1 and Δv_2 below.



2

Remember that both fall with acceleration g . The only difference is that the groom started a little later. Many people find the separation gap hardest to think about. Remember that the area under the speed versus time curve is how far the person has fallen. What's the difference in this area between the bride and groom at short times? What is it at long times? The difference at one particular time is shown below.



If the difference in areas under the speed curve increases, the separation gap increases. If the difference is constant, the separation gap is constant.

Your Response(s):

- ☺ Increased: separation
- Stayed the Same: speed gap, acceleration gap, time gap
- Decreased:

Problem 6 – Correct! – Score: 7 / 7 (43636)

**Problem:**[Report Error](#)

This problem gives you some more practice conceptualizing what acceleration is, as used in physics, and interpreting it in an everyday scenario.

This problem isn't focused on positive and negative aspects, just how large the acceleration is. That's why it asks about the magnitude of acceleration.

In baseball, a "pitcher" throws a ball, which is caught by a "catcher."



Howcast, <https://www.youtube.com/watch?v=iuvoRWnSv7E>

The pitcher's throw looks like this in slow motion:



The catcher's catch looks like this in slow motion:



Highlight House, <https://www.youtube.com/shorts/QIWHIXZDEaQ>

(Both these videos are slow motion, but not necessarily slowed down by the same factor.)

When does the ball have a greater magnitude of average acceleration: when the pitcher is pitching it or when the catcher is catching it?

- a) when the pitcher is throwing it
- b) when the catcher is catching it
- c) it's the same, or almost the same
- d) it depends on the exact pitcher and catcher

Solution:

The acceleration is how quickly the velocity changes, represented by $a_{\text{avg}} = \frac{\Delta v}{\Delta t}$.

Hidden Content

The ball may speed up or slow down a little bit while traveling from the pitcher to the catcher, but not a dramatic amount. Although Δv may not be exactly the same when the pitcher throws the ball and when the catcher catches it, it will only be different by a moderate amount.

On the other hand, when the pitcher throws the ball, their arm moves over a long distance of around two meters as they speed the ball up. This takes a time $\Delta t_{\text{pitcher}}$. We don't know exactly how long a time this is, but it's long enough for the ball to travel that entire distance.

When the catcher catches the ball, it only moves a few centimeters before stopping. Since it's moving a short distance, it won't take a long time $\Delta t_{\text{catcher}}$ to stop, so

$$\Delta t_{\text{pitcher}} \gg \Delta t_{\text{catcher}}.$$

(The symbol $\gg$ means "is a lot greater than.")

Since Δt is much bigger for the pitcher, the acceleration $a = \frac{\Delta v}{\Delta t}$ is much larger for the catcher.

If this is surprising, notice that the catcher needs to wear a glove to protect their hand; the pitcher does not. The catcher wouldn't be able to catch the ball with a bare hand. The catcher is causing so much acceleration that the ball would likely injure them if caught with a bare hand. The pitcher causes much less acceleration, and their hand is fine throwing the ball unprotected. We'll study the relationship between acceleration and *force*, a quantity more closely related to what your hand would feel, later in the course.

Hint(s):

Remember that $a = \frac{\Delta v}{\Delta t}$ when there is no change in direction during the time measured. Δv is nearly the same for the pitcher and catcher. The important factor is whether Δt is larger for the pitcher or catcher. To decide this, think about how far the ball travels while being thrown by the pitcher and how far it travels while being caught by the catcher.

Your Response(s):

☺ b) when the catcher is catching it

Problem 7 – Correct! – Score: 7 / 7 (43641)

**Problem:**[Report Error](#)

This question is quantitative, but doesn't ask you to use any formulas. Instead, you should think about how accelerations are measured and calculated and answer this question based on that process, rather than by plugging any particular numbers into a formula.

Suppose you are watching a video of an object undergoing constant acceleration. The video is playing back in slow motion, but you didn't know that. You can make accurate distance measurements, but all of your time measurements are wrong because the video is playing back at $\frac{1}{2}$ speed, but you thought it was playing at normal speed.

Suppose the real magnitude of acceleration of the object being filmed was a . What magnitude of acceleration will you find after making and analyzing your measurements?

You should put a number in the box below. For example, if the magnitude of acceleration you find is ten times the real magnitude of acceleration the object had, you should put 10 in the box.

magnitude of acceleration you find =

Solution:

The measured acceleration is .

The magnitude of acceleration is $a = \frac{\Delta v}{\Delta t}$.

Since the video is playing at half speed, everything is going half its true velocity, and the Δv you find is cut in half compared to the real Δv .

Additionally, since the video is going at half speed, all velocity changes take twice as long as they did in real life. That means your measured Δt will double.

As a result, the acceleration you find is

$$\begin{aligned} a_{\text{measured}} &= \frac{\Delta v/2}{2\Delta t} \\ &= \frac{1}{4}a \\ &= 0.25a. \end{aligned}$$

Hint(s): The answer should be less than one, since the acceleration in the video is less than real life acceleration. Since the video is running at 1/2 speed, all velocities you measure using the video will be half of their real value. Additionally, when the velocity changes, the time it takes to change will be twice as long in the video as in real life. You'll need to account for both of these effects.

Your Response(s):

☺ magnitude of acceleration you find = $1/4a$

Problem:[Report Error](#)

In this multi-part problem, you'll build a model to understand the relative importance between fast reaction time and safe speed when driving. The first part guides you on how to get a model ready for analysis, and the second part asks the question we're interested in about driving.

The difficulty of the problem set is starting to increase around here, so expect to think longer and make some false starts on the later problems. The harder you work on a problem, the better it feels to finally solve it. Remember that after you've given a problem some thought, the message boards are there for discussion.

A car is moving at velocity v_{x0} along the positive x direction. The driver needs to stop suddenly.

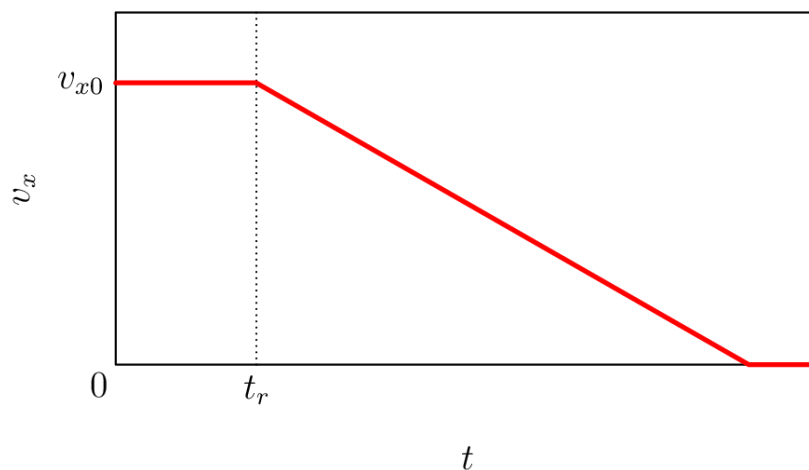
Then:

- 1) The car will continue at its original velocity v_{x0} for a short time t_r until the driver hits the brakes. t_r is the driver's reaction time.
- 2) The car will get slower and slower as the driver presses the brakes. The acceleration is a_x . Eventually the car stops.

During the second phase, constant acceleration is a fairly good model for most cars.

Note that $v_{x0} > 0$ and $a_x < 0$ due to the directions of motion and acceleration.

Below is a plot of v_x , over time in this model, beginning at the moment the driver sees a problem and ending shortly after the car stops.

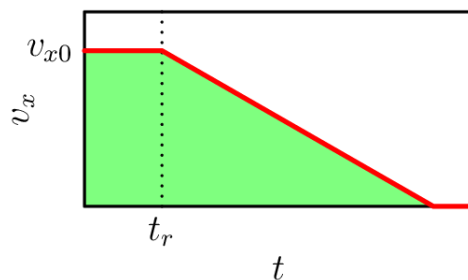


Find the distance Δx the car moves before stopping in terms of v_{x0} , t_r , and a_x .

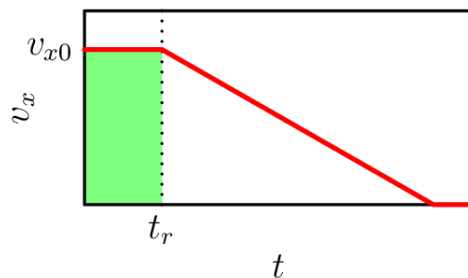
$$\Delta x = \left(v_{x0} \cdot t_r \right) - \left(\frac{v_{x0}^2}{2 \cdot a_x} \right)$$

Solution:

The displacement is the area between the curve and the horizontal axis.

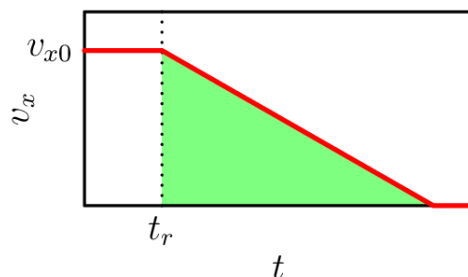


We can break this into two pieces. The first is a rectangle.



The area of this rectangle is $t_r \cdot v_{x0}$.

The next piece is a triangle.



Its area can be found from the formula $A = \frac{1}{2} \text{base} \cdot \text{height}$. The height is v_{x0} , but we need to find the base.

The slope of the hypotenuse of this triangle is a_x , so the rise is

$$-v_{x0} = a_x t_{\text{brake}},$$

where t_{brake} is the time the brakes are engaged, the base of the triangle.

That gives us

$$\begin{aligned} \text{base} &= t_{\text{brake}} \\ &= \frac{-v_{x0}}{a_x}. \end{aligned}$$

The area of the triangle is therefore

$$\frac{1}{2} \cdot \left(\frac{-v_{x0}}{a_x} \right) \cdot v_{x0} = -\frac{v_{x0}^2}{2a_x}.$$

This is a positive quantity despite the minus sign, since $a_x < 0$.

The total area is Δx , the sum of these two individual areas:

$$\Delta x = v_{x0}t_r + \left(-\frac{v_{x0}^2}{2a_x}\right).$$

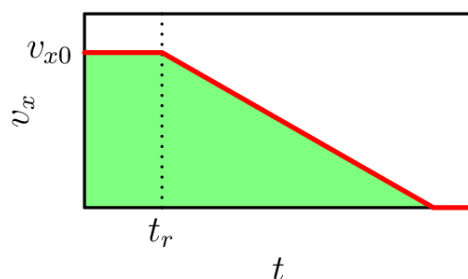
We rewrite this as

$$\Delta x = v_{x0} \cdot t_r - \frac{v_{x0}^2}{2a_x}.$$

Hint(s): 1

The displacement is the area between the curve and the horizontal axis.

2



3

Break this area into two parts and find the area of each. The first part corresponds to the first term of the answer you'll enter. The second part corresponds to the second term. To find the area of the triangle, you'll need to calculate how long it takes the car to stop, once the driver hits the brakes. To do that, remember that the slope of the velocity versus time plot is the acceleration. Also, be careful about the sign of the acceleration.

Your Response(s):

$v_{x0}, t_r, -, v_{x0}, 2, 2, a_x$

Problem 8, Part B (43644)



Problem:

[Report Error](#)

Young people generally have somewhat faster reaction times than elderly people.

Suppose that an elderly driver drives at $v_{x0;\text{elderly}} = 25 \text{ m/s}$ (about 56 mph) and the distance they need to come to a complete stop is $\Delta x = 60 \text{ m}$.

A cocky young driver thinks their faster reaction time allows them to drive safely at a higher speed. Assume the reaction times are $t_{r;\text{elderly}} = 0.6 \text{ s}$ for the elderly driver and $t_{r;\text{young}} = 0.4 \text{ s}$ for the young driver, and that a_x is the same for each driver.

The young driver needs to stop in the same distance, $\Delta x = 60 \text{ m}$, as the elderly driver. What is the maximum possible initial speed $v_{x0;\text{young}}$ for the young driver such that they can still stop within a distance Δx ?

Use the model you developed in the previous question to answer this one. Your answer should be accurate to within 0.5 m/s .

note on problem solving

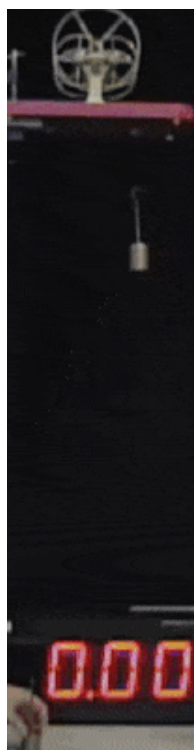
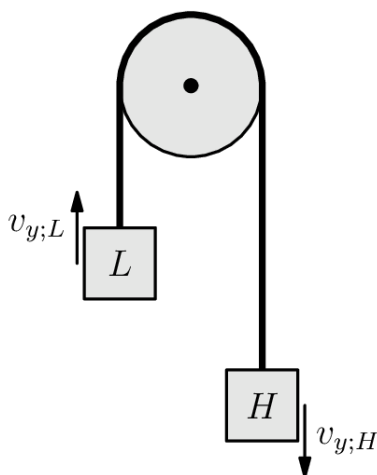
$v_{x0;\text{young}} = \text{[]} \text{ m/s}$

Problem 9, Part A (43646)

**Problem:**[Report Error](#)

In this multi-part problem, you'll practice several stages of conducting an experiment, including understanding what the experiment is for, building a model of what happens in the experiment, deciding what data to collect, deciding how to analyze that data, and drawing conclusions from the experiment.

An *Atwood machine* is a rope with weights on either end. The rope hangs over a pulley. The weights usually have different mass, and the more massive weight accelerates towards the ground while the less massive weight accelerates upward.



"Low Friction Atwood Machine," mittechtv, YouTube

You can assume the weights move with constant acceleration and that they are released from rest.

We can use an Atwood machine to measure g , assuming we have a model of the acceleration of the machine that includes g . Later, we will build such a model, but we'll ignore complications like the weight of the string or friction in the pulley.

A different experiment to measure g is simply to drop an object and time how long it takes to fall. There are several differences between this free fall experiment and an experiment using the Atwood machine.

Most of these differences make the Atwood experiment more accurate than the free fall experiment, but one does not. **Select the one**

difference that makes the Atwood machine *less* accurate than a similar free fall experiment.

You can assume that in both experiments, the times are being taken by a single person with a stopwatch.

- a) There is friction in the pulley of the Atwood machine, but negligible friction in free fall (because air drag is very small). Assume our models do not account for friction.
- b) The Atwood machine has a much lower acceleration than free fall.
- c) We can take data with many different sets of weights in the Atwood machine. This gives different accelerations. In free fall, all weights have the same acceleration.
- d) In the Atwood machine, we can release the lighter weight from ground level and find the time for the heavier weight to reach the ground. This makes the starting and ending points we're timing right next to each other, rather than far apart as for free fall.

Problem 9, Part B (43647)



Problem:

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Before we try an experiment with the Atwood machine, we should understand how we expect it to behave. A good way to start building our intuition for it is to think about extreme cases, or special cases. These are special values of the weights such that the behavior of the Atwood machine is easier to understand.

There are two extreme / special cases we should check: equal masses and very uneven masses.

We can call the mass of the heavier box m_H and the mass of the lighter box m_L . Then the equal masses special case is $m_H = m_L$. The very uneven masses extreme case is $m_H \gg m_L$. You can imagine that the heavier box has the mass of an entire mountain (but somehow still fits in the same size box) and the lighter box has the mass of a single flea.

What should the magnitude of acceleration of the Atwood machine be in each case?

$m_H = m_L$:

$m_H \gg m_L$:

$a = 0$

$a \approx g$

$a \gg g$

Problem 9, Part C (43648)

**Problem:**[Report Error](#)

Now we'll try to select a model for the acceleration of the Atwood machine.

In the Atwood machine, if we double m_H and also double m_L , it's as if we simply had two copies of the Atwood machine right next to each other, with the masses lightly touching. The two copies would each run with the same acceleration as each other, so doubling each mass in this way doesn't affect the acceleration.

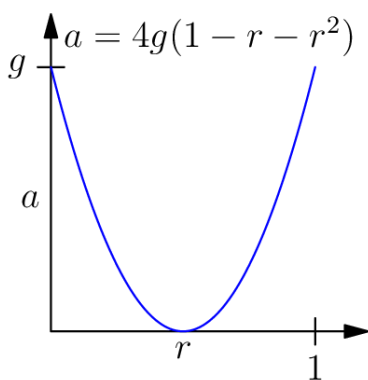
Extending this argument, it's only the ratio of masses, $\frac{m_L}{m_H}$, that affects the acceleration. We can call this ratio r .

r is always between zero and one, since by definition the denominator is bigger than the numerator.

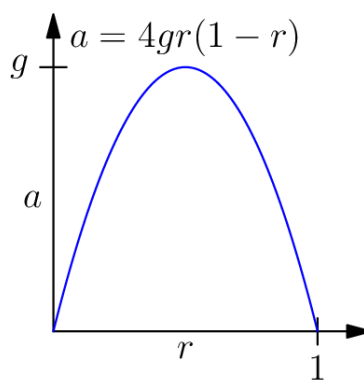
Which of the following plots is the best model for the magnitude of acceleration of the Atwood machine as a function of r ?

We've included the equations used to generate each plot as well.

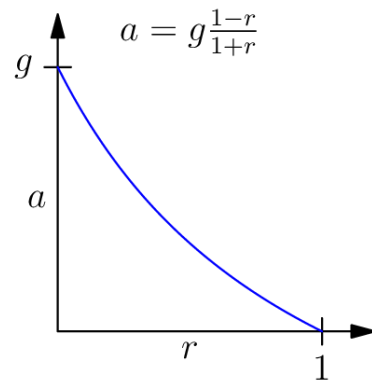
note on interpreting graphs



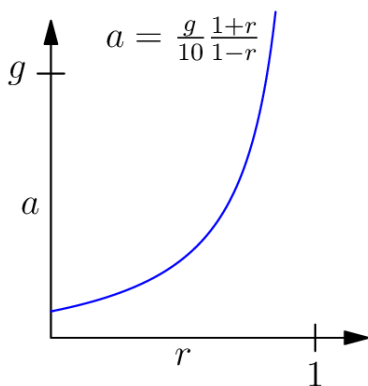
(a)



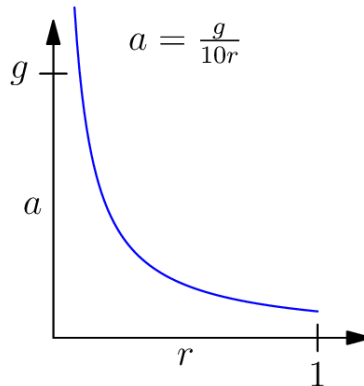
(b)



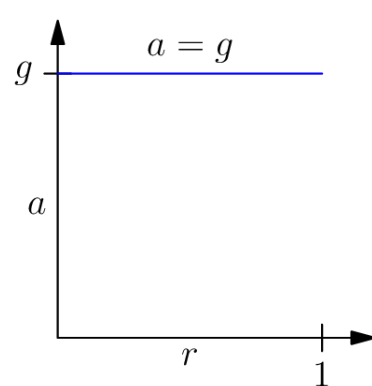
(c)



(d)



(e)



(f)

(a)

(b)

(c)

(d)

(e)

(f)

Problem 9, Part D (43649)

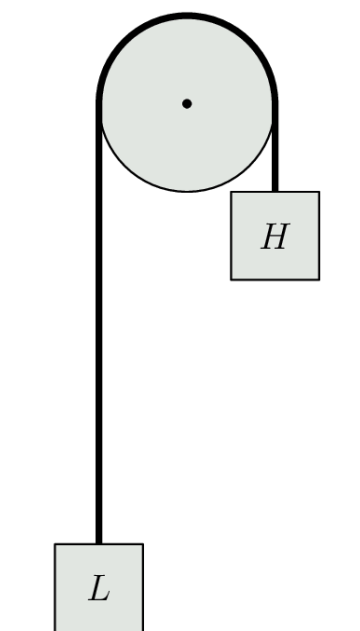
**Problem:**[Report Error](#)

We now have a theoretical model for the magnitude of acceleration a in terms of the ratio $r = \frac{m_L}{m_H}$. It's time to think about designing an experiment to measure g .

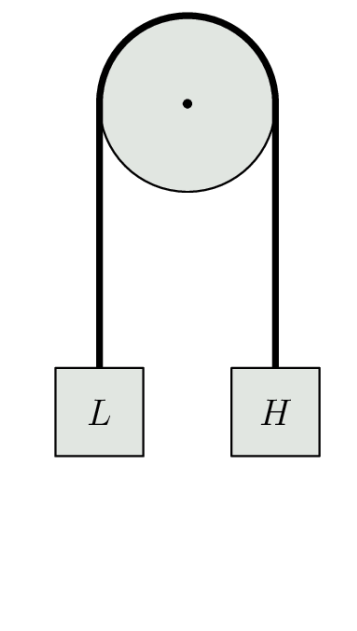
The experiment will involve running the Atwood machine and using a stopwatch.

For each of the following pairs of experimental design choices, one leads to a more accurate experiment and one leads to a less accurate experiment. Categorize each choice.

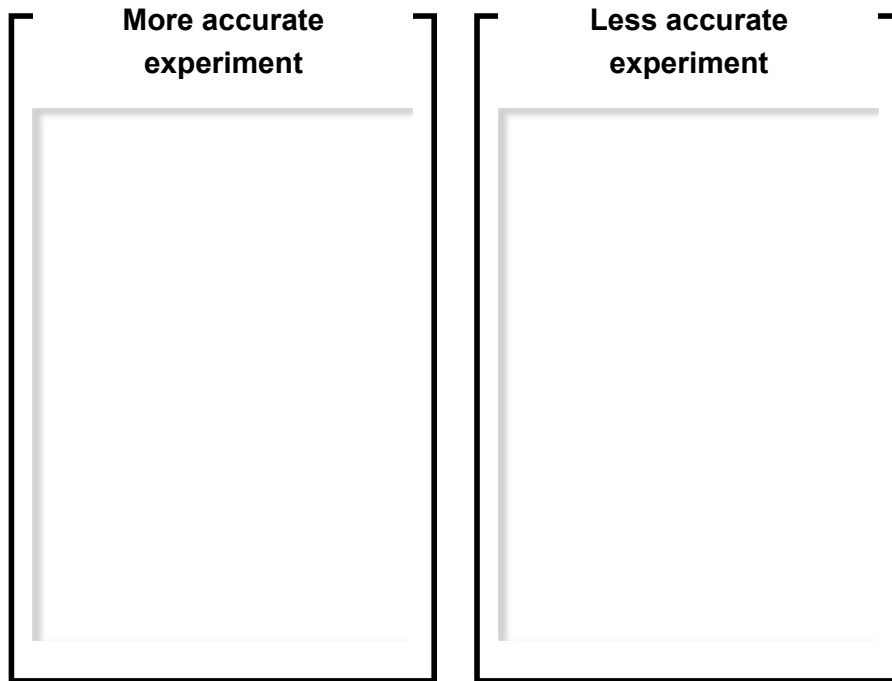
Below is an illustration of the first choice.



start heavier box at top



start heavier box half way



start heavier box half way

start heavier box at the top

use many different values of r

use a single value of r

run multiple trials for each r

run a single trial for each r

Problem 9, Part E (43705)



Problem:

[Report Error](#)

Now imagine that some scientists have conducted the experiment. For each value of r they used, they calculated an acceleration of the Atwood machine from their raw data of height and time to fall. They did this using formulas we found in class.

Then they repeated the experiment for different r values and compiled a table of r versus a .

How can they use this to calculate g ?

a) average the accelerations; the average should be g

b) plot a versus r and draw a straight line through the points, use the slope to find g

c) plot a versus the function of r found in part (C), draw a straight line through the points, use the slope to find g

d) this data can't be used to find g

Problem 9, Part F (43706)

**Problem:**[Report Error](#)

In this optional discussion problem, you conduct the experiment at home, share your results, and discuss your and other experiments with your classmates.

If you would like to do some hands on experimentation, build your own Atwood machine at home. You are welcome to buy a pulley, but you could also run some string over any smooth rod with low friction. If you can wrap a portion of a broom handle in Teflon tape, that may make a low-friction surface to support your rope. You could also try making your own pulley using instructions you find online.

A stack of coins makes a convenient way to adjust the masses in your Atwood machine.

If you collect data, find accelerations, plot them, and find g , share how you made your Atwood machine and your plot to the class forum. You can also see what other students did and compare results and techniques.

If you are having trouble finding suitable materials, you could also take data using a simulated Atwood machine, such as this one:

http://physics.bu.edu/~duffy/HTML5/Atwoods_machine.html

Discussion Problem:

This is a **Discussion** problem. Use the discuss button to visit the discussion thread for this problem and work with your classmates. You can post full or partial solutions, but please surround them in `[hide]` `[/hide]` tags as a courtesy to students who do not want the problem to be spoiled.

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